# Interview Evaluation Guidelines for Fresher Hiring

## Objective

Evaluate fresher candidates based on problem-solving ability, logical thinking, understanding of core programming concepts, real-time application knowledge, debugging and analytical skills, communication ability, and learning attitude.

## Programming Fundamentals

### Topics to Cover

- Variables, data types, operators
- Loops and conditional statements
- Functions/methods
- Arrays and strings

### Scenario-Based Evaluation Examples

- Reverse a string without using built-in functions
- Find duplicate elements in an array
- Handle invalid or null input conditions
- Find the second-largest number in an array

## Object-Oriented Programming (OOP)

### Topics to Cover

- Inheritance
- Polymorphism
- Encapsulation
- Abstraction

### Scenario-Based Evaluation Examples

- Design a simple online shopping system
- Explain inheritance using Employee and Manager classes
- Demonstrate runtime polymorphism using payment methods
- Explain data hiding using encapsulation

## Problem Solving & Debugging

### Topics to Cover

- Logic-building ability
- Optimized coding approach
- Edge case handling
- Debugging capability

### Scenario-Based Evaluation Examples

- Debug an existing code snippet
- Optimize nested loop logic
- Handle null, empty, or invalid inputs
- Identify issues in API or application flow

## Data Structures & Algorithms

### Topics to Cover

- Arrays
- Strings
- Linked Lists
- Stack & Queue
- Sorting & Searching

### Scenario-Based Evaluation Examples

- Find second-largest number in array
- Compare linear search vs binary search
- Explain use cases for stack and queue
- Explain basic time complexity using examples

## Database Knowledge

### Topics to Cover

- SQL queries
- Joins
- CRUD operations
- Normalization basics

### Scenario-Based Evaluation Examples

- Write query to fetch department-wise employee details
- Explain INNER JOIN vs LEFT JOIN
- Design simple normalized tables
- Identify duplicate records using SQL

## Web / Technology Basics

### Topics to Cover

- HTML/CSS/JavaScript
- Angular/React basics
- Java/Spring Boot basics
- REST APIs
- Git basics

### Scenario-Based Evaluation Examples

- API integration handling
- Component communication in Angular/React
- REST endpoint design basics
- Git merge conflict handling
- Form validation scenarios

## Important Interview Guidelines

- Encourage candidates to explain their thought process
- Evaluate problem-solving approach before final solution
- Prefer practical and scenario-based discussions
- Assess communication and learning attitude along with technical skills

## Resume Screening Evaluation Points

- Verify whether the technical skills mentioned in the resume match the candidate's explanation during interview
- Check project descriptions for practical technical involvement
- Evaluate understanding of technologies listed in the resume
- Ask scenario-based questions from technologies/tools mentioned by the candidate
- Validate knowledge of frameworks, APIs, databases, and programming languages claimed in the resume
- Check consistency between resume skills and coding/problem-solving capability
- Assess internship/project contributions mentioned in the resume
- Verify GitHub, certifications, or hands-on learning activities if mentioned
- Identify whether concepts are practically understood or only theoretically memorized

## Software Version Awareness – Interview Questions

### Java

- Which Java version have you worked on?
- What are the new features introduced in Java 8?
- Difference between Java 8 and Java 17?
- What is Stream API and why was it introduced?
- Scenario: Your project is migrating from Java 8 to Java 17. What compatibility issues may occur?

### Angular

- Which Angular version are you familiar with?
- Difference between AngularJS and Angular?
- What are standalone components?
- Scenario: Your Angular application is upgraded from Angular 12 to Angular 16. What areas should be validated after upgrade?

### React

- Difference between class components and functional components?
- Why were hooks introduced?
- Scenario: A legacy project uses class components. How would you modernize it using latest React features?

### Spring Boot

- Which Spring Boot version have you worked with?
- Difference between Spring and Spring Boot?
- What is auto-configuration?
- Scenario: Your Spring Boot application is upgraded to a newer version and APIs stop working. How would you debug the issue?

### Database

- Which MySQL/Oracle/PostgreSQL version are you familiar with?
- Explain compatibility issues during DB migration.
- Scenario: Database version upgrade causes query performance issues. What will you check first?

### Git & APIs

- Difference between Git merge and rebase?
- Why is API versioning important?
- Scenario: Two developers modified the same file in different branches. How will you resolve the conflict?
- Scenario: Backend APIs are upgraded while old clients still use older responses. How will you maintain compatibility?

## Optimized Interviewer Guidelines

- Validate whether resume technical skills match practical understanding
- Ask scenario-based questions from projects and technologies mentioned in resume
- Focus on fundamentals, logical thinking, and debugging ability
- Evaluate communication, learning attitude, and adaptability
- Assess real-time problem-solving instead of memorized theoretical answers
- Check awareness of software/framework versions and upgrade impacts
- Encourage candidates to explain approach before coding solutions

## Additional Evaluation Recommendation

- Questions can be twisted and scenario-driven around basic concepts to evaluate the candidate's ability to apply knowledge in real-time situations rather than relying only on memorized theoretical answers.
- Interviewers are encouraged to assess how candidates analyze problems, adapt to changing requirements, debug issues, and explain their thought process during practical discussions.